DSE Bullet Series – System Software (Solutions 04)

Name: _____ ( ) Class:____ Date: _____

1. Past Questions (2014–2022) #System or application — don't mix them up

Year	Past Question	Focus / Marks
2014	Freeware vs open source software	Both free; source code disclosed or not
2015	Shareware vs open source software	Trial limits; source code open and editable
2016	Standalone software vs cloud service	Buy + install vs subscribe + browser
2018	Two differences: Linux / macOS / Windows	Interface, compatibility, licence cost
2020	System software vs application software	Manage hardware/platform vs user tasks
2021	Why files transfer between different OS?	Common file formats and network protocols
2022	Two OS functions besides file management	Memory, process, I/O, user interface

2. Key Notes #One manages, one helps

System software manages hardware and provides a platform: operating system, device drivers and utility software. Application software helps users complete tasks: word processing, spreadsheets, browsers and accounting.

OS functions: memory, process, input/output and file management; user interface; error handling.

Device drivers let the OS communicate with specific hardware; utility software handles antivirus, backup, disk cleanup, compression and system monitoring.

Computer system types: batch processing (monthly statements); online interactive (immediate response); real-time (response within a time limit); parallel (multiple processors in one system); distributed (multiple computers at different locations share work over a network); virtualisation.
2016 DSE Q17 (Answer: A real-time system): counters at different locations are only terminals, not separate computers; seat availability must be confirmed and updated immediately. Memory tip: distributed = multiple computers across locations; parallel = multiple processors in one system.

Licensing: commercial software needs a licence; freeware does not disclose source code; open source software discloses and allows modification of source code; shareware has trial or feature limits.

Cloud vs local: cloud software needs no installation, works across devices and updates automatically; it requires a network and has higher long-term cost.
*Common mistakes: assuming ‘freeware = open source software’; confusing system and application software; writing only ‘convenient’ or ‘easy to use’; comparing without a common criterion.

3. Similar Questions #Same angle, different software

1. The school uses a cloud word-processing service, not standalone software.

(a) State **two** benefits of switching to a cloud service.

【Answer】No need to install on every computer; can be used through a browser; teachers and students can access the same file on different devices; software updates automatically, ensuring consistent versions; charges are based on the number of users, lowering initial cost.

(b) State one risk of using a cloud service.

【Answer】Requires a stable network; service is unavailable when the connection is down; long-term subscription costs may be higher; files are stored with a third party, raising privacy and data-leakage risks.

2. The technician adds application software after installing the operating system.

(a) State **two** types of application software and explain their purposes.

【Answer】Word processor: edit and format documents; spreadsheet: calculations, charts and data analysis; browser: browse web pages; presentation software: create presentations. (Any two + purpose)

(b) Explain one difference between application software and system software.

【Answer】Application software completes specific user tasks, e.g. word processing; system software manages hardware and provides a platform for application software to run, e.g. an operating system.

3. The school is considering replacing commercial software with open source software.

(a) State **two** benefits of using open source software.

【Answer】Free to use, saving licence fees; source code is open, so it can be modified to suit the school's needs; community support and updates; avoids being locked into a single vendor.

(b) Explain why ‘freeware’ is not necessarily the same as ‘open source software’.

【Answer】Freeware means only that it is free of charge; its source code may not be disclosed and users cannot modify it. Open source software discloses its source code and can be modified and redistributed.